Year 8 Science – Booklet 24: Chemistry

Compounds & Mixtures — ANSWERS

Note: this booklet's content is byte-for-byte identical to Year 8 Booklet 22 (same Compounds and Dissolving pages), so the same worked answers apply throughout.

Compounds – Quick Test (p.79)

1. An element in its atomic form – separate, unjoined atoms of one element.
2. Molecules of an element – atoms of the same element joined together in pairs.
3. A compound – atoms of two different elements chemically joined together.
4. A gas.
5. A thermometer.
6. Reactants.
7. Products.
8. The product (iron sulphide) is a black, non-magnetic solid – very different from the magnetic iron and yellow sulphur it was made from, and no longer attracted to a magnet, showing a new substance with different properties has formed.
9. Yes – a compound always has a fixed composition, e.g. iron sulphide (FeS) always has one iron atom for every one sulphur atom.
10. No – the elements in a compound are chemically joined and cannot be separated by simple physical means (unlike a mixture).

Compounds (workbook, p.78–79)

A. Multiple Choice (5 marks)

1. c) it's a yellow solid
2. b) it's a magnetic metal
3. d) it's a compound
4. a) use a magnet
5. a) it cannot be easily separated

B. 1. Anagrams and definitions (5 marks)

cduopomn → **compound** — a substance which is made of atoms of two or more elements that have been chemically joined together
yeproprt → **property** — a quality which is always present in a substance, however much of it is present
eeculomc → **molecule** — a small piece of a substance which has all the properties of the substance
mentag → **magnet** — a substance which repels another magnet
icatreno → **reaction** — a chemical change

B. 2. Complete the passage (10 marks)

Elements are made of only one type of **atom**. If atoms of the same elements are joined together they form **molecules** of the element. The oxygen and nitrogen in the air around us are in the form of molecules. If the atoms of two (or more) different elements are chemically joined together they form molecules of a **compound**. New substances are formed by chemical **reactions**. These new substances are called the **products**. The products of a chemical reaction can have very different properties from the **reactants** at the start of the reaction. When iron and sulphur are heated together they can form a new substance called **iron sulphide**. Iron is a magnetic **metal**. Sulphur is a **yellow** solid. But, **iron sulphide** has very different properties from both iron and sulphur – it is a non-magnetic black solid.

C. 1. How could Nikesh tell a chemical reaction was taking place? (1 mark)

Bubbles of gas would be seen forming in the flask, and/or the reading on the mass balance would decrease as the carbon dioxide gas produced escapes into the air.

C. 2. a) i) What else can have caused the decrease in mass? (2 marks)

The carbon dioxide gas produced by the reaction escaping from the flask into the air.

C. 2. a) ii) Name one of the reactants

Magnesium carbonate (or hydrochloric acid).

C. 2. b) Which diagram (A, B, C or D) best represents the CO■ gas produced? (1 mark)

C – a gas is represented by particles spread far apart with space between them (unlike the tightly-packed solids A and D); since CO■ is a compound, its particles are shown as joined molecules (bonded pairs), which is diagram C rather than B (single, unjoined atoms spread apart).

Dissolving – Quick Test (p.53)

1. 102 g.
2. The solute.
3. The solvent.
4. A solution.
5. Soluble.
6. It generally increases the amount that can dissolve (increases solubility).
7. A solution in which no more solute can dissolve at that temperature.

8 & 9. Reading exact values off the solubility graph isn't precise from this scan – based on the typical shape of these curves, potassium nitrate is roughly **320 g** per 100 g of water at 20°C, and sodium nitrate is roughly **880 g** per 100 g of water at 20°C. Please check these against the actual graph in your copy.

10. It increases – more can dissolve at a higher temperature.

Dissolving (workbook, p.52–53)

A. Multiple Choice (5 marks)

1. b) 105 g
2. b) 100.2 g
3. c) saturated solution
4. d) warm the solution
5. a) their solubility increases

B. True or false (7 marks)

a) Copper sulphate is insoluble in water – **False** (it is soluble)
b) Sugar is more soluble in hot water than cold water – **True**
c) Most solutes are more soluble at higher temperatures – **True**
d) When no more solute can dissolve, a saturated solution has formed – **True**
e) 2 g of solute in 100 g of solvent gives a 102 g solution – **True**
f) Water is a solvent for all solids – **False** (some solids, e.g. sand, are insoluble in water)
g) A warmed saturated solution can normally dissolve a little more solute – **True**

C. 1. Harry's copper sulphate experiment (1 g crystals + 20 g water)

a). Water.

b). Copper sulphate.

c). 1 g + 20 g = **21 g**

d). There would be no more visible solid crystals left in the beaker – the liquid would look clear/uniform with nothing settled at the bottom.

e). Stir the mixture, use warmer water, or use smaller/crushed crystals (more surface area).

f). A saturated solution.

g). Warm/heat the solution (increase the temperature), or add more water.

Note: This is a Collins “KS3 Science Success Guide” page-set (“Compounds” and “Dissolving”), reproduced here by Aristeia Academy Tuition, rather than an exam-board past paper – there is no publicly downloadable official markscheme for it. These answers were worked out directly from the notes and data given earlier in the booklet.